

SAHIL GOYAL

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 Scholar

 GitHub

 LinkedIn

 Medium

EDUCATION

Indian Institute of Technology Roorkee

Integrated M.Sc. in Applied Mathematics

Department rank 1 (Gold medal)

2019 - 2024

Overall GPA: **9.44**/10

PUBLICATIONS

1. **MaGNeTS: Masked Generative Nested Transformers with Decode Time Scaling.** [Paper] [Poster]
Sahil Goyal*, Debapriya Tula*, Gagan Jain, Pradeep Shenoy, Prateek Jain, Sujoy Paul [ICML '25]
2. **Design-o-meter: Towards Evaluating and Refining Graphic Designs.** [Paper] [Poster] [Project]
Sahil Goyal*, Abhinav Mahajan*, Swasti Mishra, Prateeksha Udhayan, Tripti Shukla, K J Joseph, Balaji Vasan Srinivasan. [WACV '25].
3. **Emotionally Enhanced Talking Face Generation.** [Paper] [Poster] [Code]
Sahil Goyal, Shagun Uppal, Sarthak Bhagat, Yi Yu, [ICCV-W '23]
Yifang Yin, Rajiv Ratn Shah. [ACM MM-W '23]
4. **Emotional talking faces: Making videos more expressive and realistic.** [Paper] [Demo]
Sahil Goyal, Shagun Uppal, Sarthak Bhagat, Dhroov Goel, Sakshat Mali, [ACM MM Asia '22]
Yi Yu, Yifang Yin, Rajiv Ratn Shah. [Best Demo paper]

EXPERIENCE

Pre-Doctoral Researcher at Google Deepmind

June 2024 - Present

Advisors: Dr. Sujoy Paul, Dr. Prateek Jain, Dr. Aditya Kusupati, Dr. Pradeep Shenoy

- Developed an **adaptive computation** method [1] for multi-step generative models using **nested transformers**.
- Achieved **~3x** reduction in inference FLOPs for image/video generation.
- Designing **efficient and elastic sampling methods** for Google's SOTA video generation model **Veo**.
- Optimizing **model compression** pipeline for **Gemini** in a compute-constrained setting.
- Devising effective **structured-pruning** and **distillation** techniques.

Research Intern at Adobe Research

May 2023 - July 2023

Advisors: Dr. Joseph KJ, Dr. Balaji Vasan Srinivasan

- A novel **self-supervised** approach [2] to evaluate **Graphic Designs** quantitatively.
- Used compact **Siamese networks** (**0.5M** params) along with Design Heuristics to score Graphic Designs.
- Tailored advanced **Genetic Algorithms** (NSGA-II) for **Layout refinement**.
- Significantly outperformed multimodal LLMs GPT-4o and LLaVA-NeXT (7B) in ranking graphic designs.
- Achieved SOTA mean-IOU of **0.54** in layout refinement on the Crello dataset.
- Deployed the entire system as a Flask Web Application with potential integration into **Adobe Express**.

Research Intern at MIDAS Lab @IIITD

Sep 2021 - Jan 2023

Advisors: Dr. Rajiv Ratn Shah

- A novel **CNN-based talking face generation** network [3, 4] involving **lipsync** and explicit **emotions**.
- New skip-connections-based concatenation technique to concatenate video, audio, and emotion embeddings.
- SOTA results on **CREMA-D** dataset (**83.2%** emotion classification accuracy and **FID** metric: **5.29**).
- Deployed the entire system as an interactive web interface.

- Optimized the 3-D space optimization (a heuristic algorithm) for transport.
- Used **vectorized operations** and **parallel computing (Numpy and PyTorch)**.
- Reduced the algorithm’s running time by **95 %** on real-world data.

SELECTED ACHIEVEMENTS

- Among **15**, from 18,000+ applicants, selected for Google’s Pre-Doctoral Researcher Program.2024
- **Department Gold medal** (IIT Roorkee - Applied Mathematics).2024
- **Best Demo Paper** at ACM MM ASIA conference.2022
- **Gold medal** at Inter IIT Tech Meet.2022
- **Open Source:** Contributions to **PyTorch-Vision**.2023
- Awarded ACM SigMM student travel grant for ACM MM Asia.2022
- JEE Advance Rank of **1583** among 1.5M students.2019
- **600+** stars on  **GitHub**.

PROJECTS B.C. (BEFORE CHATGPT)

Model Extraction Attacks for Video Classification [Code]

Feb 2022 - March 2022

- Model extraction on hierarchical transformer (**Swin-T**) and 3D-CNN (**MoViNet**) video classification models.
- Conducted extraction across Grey-box (5% original data) and Black-box (synthetic data only) settings.
- Used novel data augmentation, knowledge distillation and synthetic data generation techniques.
- Achieved **54.8%** top-5 classification accuracy with **R(2+1)D** as clone model.

Global-Wheat-Detection-2021-Challenge [Code]

June 2021 - July 2021

- Performed detailed **EDA** on wheat images acquired from different continents.
- Experimented **Faster RCNN** with different **ResNet** and ResNext backbones for **wheat-head detection**.
- Utilized Data augmentation, Non-Maximum Suppression, Weighted boxes fusion, and Pseudo Labeling.
- Achieved **56.2%** Average Domain Accuracy.

Multimodal Image-to-Image Translation [Code] [Blog]

Dec 2020

- Implemented **BicycleGAN** from scratch in PyTorch.
- Explored **Edges-to-Images** (Edges to Shoes dataset) and **Denoising images** (CIFAR10 dataset) tasks.
- Achieved diversity-based **LPIPS** score of **0.16** (Edges to Images) and **0.33** (Denoising images)
- Validated **results** by closely approaching the paper’s real images **LPIPS** benchmark (**0.262**).

POSITIONS OF RESPONSIBILITIES

- **Reviewer** at CVPR, ICML, ICLR, ICCV, AAMAS, WACV conferences.2024 - Present
- **Senior member** at **Data Science Group IITR**2020 - 2024
- **Undergraduate Teaching Assistant** at Academic Reinforcement Program IITR2021
- **Teaching Volunteer** at National Social Service IITR2019 - 2020

RELEVANT KEY COURSES

Machine Learning	Stanford CS229 (Machine Learning), Stanford CS231n (Computer Vision), Deep Learning Specialization, Data Mining, CVIT Summer School of AI.
CS and Maths	Data Structures and Algorithms, Linear Algebra, Mathematical Statistics, Graph Theory, Statistical Inference, Parallel Computing, Discrete Mathematics, Numerical Methods, Real Analysis, Functional Analysis, Topology, Combinatorics, Differential Equations

RESEARCH INTERESTS

Efficient ML, Generative Models, Computer Vision, Adaptive Computation, Knowledge Distillation